Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, adherent

Species Used Hamster, CHO, ovary

Molecules Electroporated Proteins: restriction enzymes (*Eco* RI, *Sca*I, *Dra* I) and catalase

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Before the Pulse

Cell Growth Medium McCoys 5a (GIBCO/BRL, Sigma)

Growth Phase at Harvest Exponential

Pre-pulse Incubation Hepes Buffered Saline (HBS)

Wash Solution Phosphate Buffered Saline (PBS) or serum-free medium

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature 4°C

Electroporation Medium* HEPES Buffered Saline (HBS)

Cell Density 2 x 10 (6) cells / ml

Volume of Cells 0.8 ml

DNA Concentration Not given

DNA Resuspension Buffer Not given

Volume of DNA Not given

Cuvette Gap 0.4 cm

Voltage 0.3 kV

Field Strength 0.750 kV/cm

Capacitor 960 µF

Resistor (Pulse Controller) Ω none

Time Constant 16 to 22 msec

After the Pulse

Outgrowth Medium McCoys 5A

Outgrowth Temperature 37 °C

Length of Incubation 20 hr.

Selection Method or Assay Used Not given

Electroporation Efficiency Not given

Per Cent Survival 40 to 60%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
Cortez, F., and Ortiz,T.*Mutation Research* **246**(1):221-6
PBS: 1x = 8g NaCl,0.2g KCl,0.2g KH2PO4, 1.15g Na2HPO4
HBS: 10mM HEPES,pH 7.2,150 mM NaCl, 5 mM CaCl2

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Survey Number 090